Chapter 3 exercises

DS03-E01 • Circular wraparound

> Use head and size with capacity C=4. From the empty state `(head,size)=(0,0)`, enqueue A,B,C, dequeue twice, enqueue D,E,F, then dequeue three times. Give head, size, logical sequence, and physical positions after every operation.

DS03-E02 • Expression conversion

> Convert `(9-4)*(2+3)` to prefix and postfix. Draw its expression tree and evaluate it with a stack. Explain why the pop order for postfix `8 3 -` cannot be exchanged arbitrarily.

DS03-E03 • The first DFS path

> Design an unweighted graph of at most six vertices and a neighbor order such that DFS first reaches the target along a longer path than BFS finds. List both visit orders and predecessor assignments.
>
> Hint. Give the start both a short path and a longer path that DFS explores first.

DS03-E04 • Syntax and safe evaluation

> At what conditions must postfix inputs `2 +`, `2 3`, and `8 0 /` be rejected? For signed 64-bit values, what arithmetic checks are needed besides division by zero? A complete lexer is not required.

DS03-E05 • End-of-input and grammar

> Convert `9-4*2/1+3`, listing every token, operator stack, and output, including EOF. Give rejection points for `(9-4`, `9-4)`, and `9-`. Compare postfix forms and values for left- and right-associative `2^3^2`.

DS03-E06 • Queue growth and the abstract value

> Capacity is 4 with physical=[E,F,C,D],head=2,size=4. Grow to 8. List old and new slots for every logical value and the committed head/size. Refute direct slot copying with retained head using an actual accessed position. Analyze one growth and repeated doubling.

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DS03-E07 • Mixed operations with two stacks

> Use the chapter's I,O algorithm. Starting empty, enqueue 4, enqueue 7, dequeue, enqueue 9, dequeue, dequeue. After each step give I and O bottom-to-top, the abstract FIFO sequence, and the exact number of stack pushes/pops. Compute amortized costs with Φ=2|I|. Do not transfer while O is nonempty.

DS03-E08 • Stack permutations and required capacity

> Fixed input is 1,2,3,4 with no bypass. Test outputs 2,1,4,3 and 3,1,4,2. For a feasible output, list operations and the minimum capacity; otherwise identify the first blocked requested value. Use the recurrence to count all feasible outputs and explain why the count is not 4!.

DS03-E09 • Bracket kinds and order

Trace the stack or first rejection for `([])`, `([)]`, `(()`, and the empty string. Why is counting opening and closing brackets of each kind insufficient?

DS03-E10 • Palindrome with known length

Trace the first-half stack method on `racecar` and `abca`. State how to handle the middle symbol for odd length and prove that at most $\lfloor n/2 \rfloor$ comparisons are needed. Does testing UTF-8 bytes always agree with testing code points?

DS03-E11 • Boundaries of $a^n b^n$

Determine whether `aabb`, `aab`, `abb`, `abab`, and the empty string belong to $\{a^n b^n \mid n \geq 1\}$. Explain each acceptance or rejection. Which result changes for $n \geq 0$?

DS03-E12 • Circular queue capacities

For an array of length 4, compare head/size, a reserved-slot front/rear representation, and front/rear with a tag. Give maximum occupancy and empty/full tests, and explain why the tests cannot be interchanged.

References and attribution

Lin Cing, handwritten data structures notes; C++ draft stack and queue; MIT 6.006, Lecture Notes 9–10.

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